

Answer Key — Biology Grade 11

Sub-Strand 2.3: Excretion in Plants • Lesson 3: Where Do the Wastes Go? Shedding Old Leaves and Locking Away Crystals

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	D	C	B	C	A	A

USE AFTER: PREDICT PHASE — Predict: hardened Acacia gum and yellow leaf prediction (Think-Pair-Share)

Q1. Plants have no kidneys. A student predicts where a plant keeps wastes that cannot dissolve in water, such as calcium oxalate crystals, tannins and resins. Which prediction fits what the lesson develops?

- A. They dissolve in water and leave the plant as sweat through the leaves
- B. They are stored in the growing tips so the plant can reuse them quickly
- C. Plants do not make any wastes, so there is nothing to store
- D. They are stored in non-vital parts such as bark, old leaves, fruits and flowers, which are later shed**

Correct answer: D. Insoluble wastes are deposited in non-vital organs that the plant can later shed. The other choices are misconceptions: all waste leaves as liquid, wastes sit in vital tissue, or plants have no waste.

USE AFTER: OBSERVE PHASE — Part C: Microscopy, crystal count table for young and old leaves

Q2. A group examines slides of a young green mango leaf and an old yellow mango leaf under a light microscope. They count needle-shaped crystals in 3 fields of view for each leaf. Which result would support the idea that old leaves store wastes?

- A. The same number of crystals in both leaves, as all leaves hold equal wastes
- B. Crystals only in the veins of the young leaf
- C. More crystals in the young green leaf than in the old yellow leaf
- D. More crystals in the old yellow leaf than in the young green leaf**

Correct answer: D. Old, non-vital leaves accumulate stored insoluble wastes, so more crystals are expected in the old yellow leaf than in the young leaf.

USE AFTER: OBSERVE PHASE — Part C: Microscopy, iodine test on a second slide

Q3. In the leaf microscopy practical, learners add a drop of iodine to a second slide of leaf tissue. Why do they do this?

- A. To make the leaf tissue produce more crystals
- B. To dissolve calcium oxalate crystals so they can be counted
- C. To check for starch, which stains blue-black, so it is not mistaken for crystals**
- D. To kill the cells so the crystals become bigger

Correct answer: C. Iodine stains starch blue-black, which helps learners tell starch grains apart from crystals.

USE AFTER: OBSERVE PHASE — Part C: Microscopy, dilute acid test on a third slide

Q4. A learner sees needle-shaped crystals in old leaf tissue. After adding a drop of ethanoic acid at the edge of the coverslip, the crystals do not dissolve. Which conclusion is best supported?

- A. The crystals are probably sugar, because sugar never dissolves
- B. The crystals are probably calcium oxalate, which does not dissolve in ethanoic acid**
- C. The crystals are probably starch, because they stain blue-black
- D. The crystals are probably calcium carbonate, because they fizz in acid

Correct answer: B. Calcium oxalate crystals do not dissolve in ethanoic acid, whereas carbonate crystals fizz or dissolve in dilute HCl. Starch is identified with iodine, not acid.

USE AFTER: OBSERVE PHASE — Part C: Microscopy, groups who see no crystals

Q5. One group looks carefully at slides of an old leaf and finds no crystals in any field of view. What should the group do?

- A. Draw crystals anyway so their result matches the other groups
- B. Throw away the slide and copy another group's results
- C. Report honestly that no crystals were seen, record it as data, and look in other places such as the veins or the leaf edge**
- D. Decide that the old leaf stores no wastes at all

Correct answer: C. Honest reporting is essential, and 'no crystals' is still data. One negative slide does not show that the leaf has no stored wastes, so the group should look in other parts of the leaf.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning (CER) statement on how plants get rid of insoluble wastes

Q6. Groups find more calcium oxalate crystals in old yellow leaves than in young leaves. Which reasoning best explains why leaf fall counts as excretion?

- A. Wastes are locked in a harmless solid form in tissue the plant no longer needs, and they are removed when the leaf falls**
- B. Leaf fall removes crystals from the young leaves, which are the main storage site
- C. The crystals dissolve in the yellow leaf and leave as droplets at the leaf edges
- D. Leaf fall only helps the plant to lose water vapour through stomata

Correct answer: A. Insoluble wastes are deposited as crystals in old, non-vital leaves. They do not harm living cells, and the plant removes them when the organ is shed, so no excretory organ is needed.

USE AFTER: MODEL BUILDING PHASE — Revising the class model: adding deposition and tissue fall alongside diffusion, transpiration and guttation

Q7. Learners revise the class model of plant excretion. Which pair correctly matches a waste with how it is removed?

- A. Calcium oxalate crystals stored in old leaves: removed when the leaves are shed**
- B. Droplets at leaf edges (guttation): deposited as crystals in bark
- C. Water vapour: removed when old leaves fall
- D. Resin: breathed out through the stomata as a gas

Correct answer: A. Insoluble wastes such as calcium oxalate are deposited in non-vital organs and leave with them when shed. Water vapour leaves by transpiration, and guttation releases liquid droplets, not crystals.